AI + IoT Smart Bio-Security Home System

1. Project Title

AI and IoT-Based Smart Bio-Security Home System for Intelligent Physical, Environmental, and Biological Monitoring

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2. Project Overview

This project proposes an intelligent IoT-based home security system that combines **Artificial Intelligence, computer vision, environmental monitoring, and molecular biology** into one integrated platform.

Unlike conventional smart-home security systems that rely heavily on individual motion, door, and window sensors, the proposed system will use **camera-based Artificial Intelligence** to identify people, detect movement, recognize door states, and identify potentially unauthorized activity.

An ESP32-CAM or similar camera device will provide visual data, while an AI model will analyze the camera stream to detect events such as:

- Human presence
- Movement
- Entry through a monitored door
- Door opening/closing
- Potential unauthorized access

Environmental sensors will monitor conditions such as temperature, humidity, and air quality.

The project's distinctive component will be a **molecular biology-based biological monitoring subsystem**. Air samples can be collected and analyzed for a selected biological target, such as an indoor fungal target, using an appropriate molecular technique. The resulting biological information will then be incorporated into the IoT platform.

The complete system will therefore combine:

AI + Computer Vision + IoT + Smart Home Security + Molecular Biology

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3. Problem Statement

Traditional smart-home security systems commonly use separate sensors for doors, windows, motion, and other events.

Although these systems are effective for basic detection, they provide limited contextual information.

For example, a motion sensor can detect movement but cannot determine whether the movement was caused by:

- A person
- A pet
- Another object

Similarly, a magnetic door sensor can detect that a door opened but cannot determine:

- Who opened it
- Whether someone entered
- Whether the person was authorized
- What happened immediately before or after the door opened

The proposed system addresses this limitation by using **AI-based computer vision** to interpret camera data and provide more contextual information.

At the same time, the project introduces molecular biological monitoring to investigate potential biological contamination in indoor environments.

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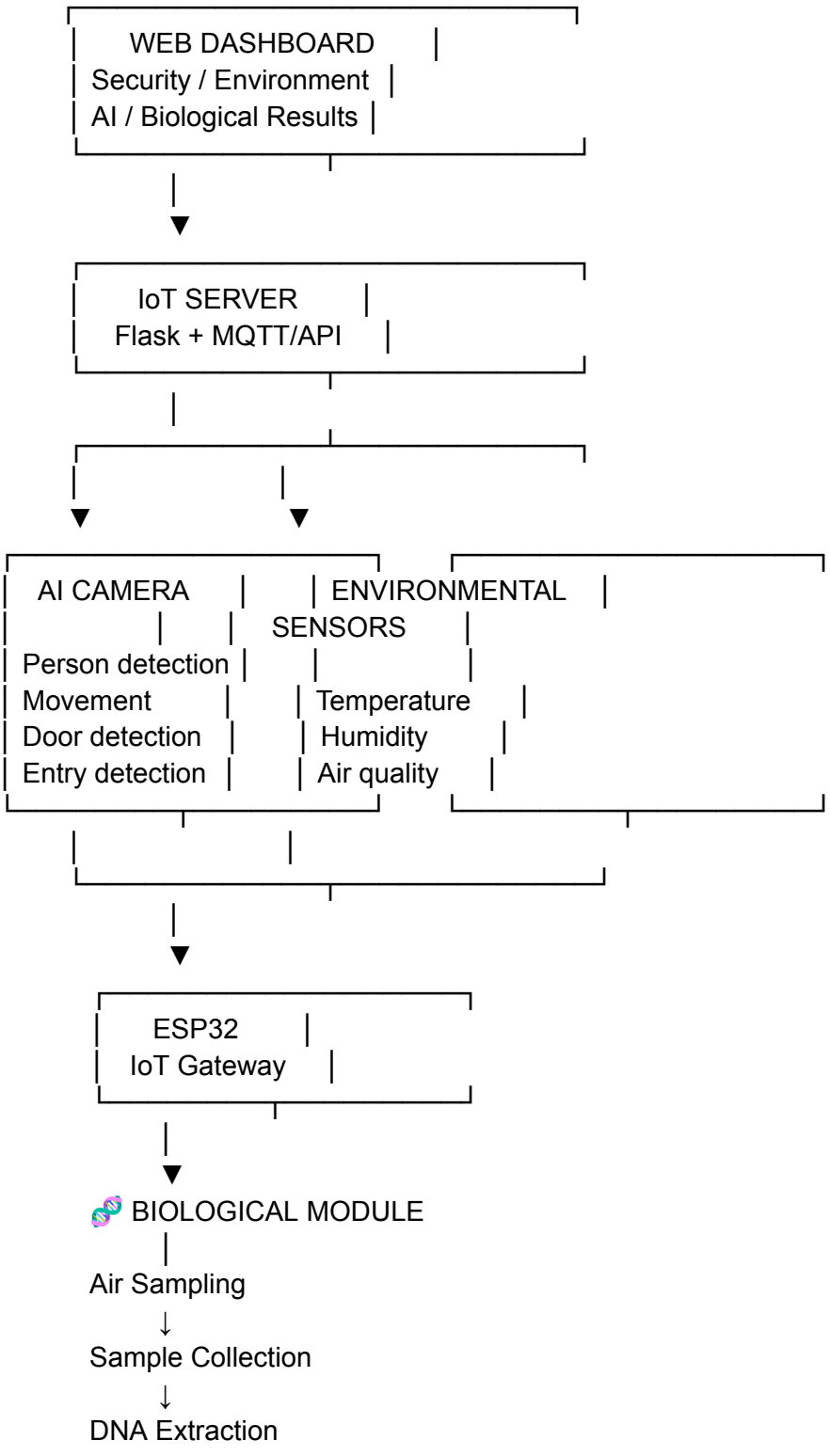
4. Main Objectives

The project aims to:

1. Develop an IoT-enabled smart-home security prototype.
2. Replace conventional motion detection with AI-based computer vision.
3. Detect people and movement using cameras.
4. Detect door opening and closing using computer vision.
5. Identify potentially unauthorized entry.
6. Monitor environmental conditions.
7. Develop a biological sampling and molecular detection workflow.
8. Connect AI, sensors, and biological results through an IoT platform.
9. Develop a web-based monitoring dashboard.
10. Generate real-time alerts.
11. Store historical security, environmental, and biological data.
12. Demonstrate an interdisciplinary system combining Computer Engineering, Software Engineering, and Molecular Biology.

5. System Architecture

The proposed architecture is:



↓
PCR / LAMP / Other Assay
↓
Biological Result
↓
IoT System

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6. AI-Based Security System

The camera will serve as the primary physical-security detection mechanism.

6.1 Person Detection

The AI system will analyze the camera feed to determine whether a person is present.

Possible technologies include:

- YOLO
- MobileNet-SSD
- TensorFlow Lite
- OpenCV
- Edge Impulse

For example:

Camera
↓
AI Model
↓
Person detected
↓
Is security system armed?
↓
YES
↓
🚨 Unauthorized movement detected
↓
IoT Server
↓
Phone/Web notification

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7. AI-Based Movement Detection

Instead of using a PIR motion sensor, the camera will determine whether meaningful movement is occurring.

The system can use:

- Object detection
- Object tracking
- Frame comparison
- Optical flow
- Background subtraction

The system can distinguish between a static environment and a moving person.

For example:

Frame 1 → Empty room
Frame 2 → Person appears
Frame 3 → Person moves
Frame 4 → Person approaches door

The AI system can combine these events to determine that a person is moving through the monitored area.

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8. AI-Based Door Detection

A dedicated magnetic door sensor will not be required for the main prototype.

Instead, the camera will monitor a predefined **Region of Interest (ROI)** containing the door.

The AI/computer-vision system will determine whether the door is:

CLOSED
↓
Door movement detected
↓
Door OPEN
↓
Person detected
↓
Potential entry

This can be implemented using computer vision techniques such as:

- Door-state classification
- Object detection
- Image classification
- Frame comparison
- ROI analysis

A more advanced version could use an AI model trained specifically on images of the prototype door in different states.

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9. Intelligent Intrusion Detection

The most important advantage of combining AI detection methods is that the system can make decisions based on **multiple observations**.

For example:

Door opens
+
Person detected
+
Person enters restricted zone
+
Security system armed
↓
🚨 POSSIBLE INTRUSION

Rather than simply saying:

"Motion detected"

the system could report:

🚨 Possible unauthorized entry detected at the front door.

The event can include:

- Timestamp
- Camera location
- Detection type
- Screenshot
- Confidence score
- System status

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10. Optional Face Recognition

Face recognition can be considered as an optional advanced feature.

The system could distinguish between:

Known person

→ Normal event

and

Unknown person

→ Potential security alert

However, face recognition should be treated separately from basic person detection because it introduces additional privacy and accuracy considerations.

The core project will remain functional without it.

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11. Environmental Monitoring

Environmental sensors will remain in the system, but their role will be focused on **environmental monitoring rather than security-event detection**.

Possible measurements include:

- Temperature
- Humidity
- Air quality
- Gas concentration
- Smoke-related conditions

For example:

Temperature: 25.1°C
Humidity: 72%
Air quality: Moderate

The system can analyze these values and identify unusual environmental conditions.

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12. Biological Monitoring Module

The biological module is the feature that makes the project interdisciplinary.

The team will select an appropriate indoor biological target, potentially an indoor fungal target.

A possible workflow is:

Indoor Air
↓
Air Sampling
↓
Collection Filter
↓
Sample Preparation
↓
DNA Extraction
↓
Target-Specific Molecular Assay
↓
Positive / Negative Result
↓
IoT Platform

The Molecular Biology team member will investigate:

- Target organism
- Sampling method
- DNA extraction
- Primer selection
- Molecular assay
- Positive and negative controls
- Interpretation of results

The biological test will be treated as a **research prototype**, rather than a clinically validated diagnostic system.

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13. AI + Environmental + Biological Intelligence

An important feature of the project will be combining different types of information.

For example:

Humidity ↑
+
Temperature suitable for fungal growth
+
Environmental conditions persist
↓
AI/Software system identifies
potential biological-risk conditions
↓
Molecular sample requested
↓
Molecular analysis
↓
Target detected / not detected

This allows the system to distinguish between:

Environmental risk

and

Actual molecular detection

Humidity alone will not be interpreted as proof of biological contamination.

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14. IoT Architecture

The system will use ESP32-based hardware to connect the physical environment with the software platform.

Possible architecture:

Camera
↓
ESP32 / Processing Device
↓
Wi-Fi
↓
MQTT / REST API
↓
Flask Backend

↓
Database
↓
Web Dashboard

MQTT can be used for real-time communication between IoT devices and the server.

Example topics:

home/camera/person
home/camera/movement
home/door/status
home/security/alert
home/environment/temperature
home/environment/humidity
home/environment/airquality
home/biology/result

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15. Web Dashboard

The dashboard will provide a centralized view of the home.

Example:

```
╔═══════════════════════════════════════╗
║        SMART BIO-SECURITY             ║
╠═══════════════════════════════════════╣
║ Security:      🟢 ARMED               ║
║ Person:        🔴 DETECTED            ║
║ Door:          🟠 OPEN                ║
║ Movement:      🔴 DETECTED            ║
║                                       ║
║ Temperature:   24.3°C                 ║
║ Humidity:      68%                    ║
║ Air Quality:   🟢 NORMAL              ║
║                                       ║
║ Biology:       🟢 NO TARGET DETECTED  ║
╚═══════════════════════════════════════╝
```

The dashboard can also display:

- Live camera feed
- AI detection boxes

- Event history
- Environmental graphs
- Biological test history
- Alarm status
- Device status

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16. Alert System

The system will generate alerts based on detected events.

Security Alert

> 🚨 Possible unauthorized entry detected.

Person Detection

> 👤 Person detected in restricted area.

Door Detection

> 🚪 Front door opened.

Environmental Alert

> ⚠️ Abnormally high humidity detected.

Biological Alert

> 🧬 Biological target detected in analyzed sample.

Alerts can be displayed through the web application and, if implemented, sent through external notification services.

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17. Hardware Components

The revised prototype will require fewer physical security sensors because AI will replace the main door and motion sensors.

Component	Purpose
ESP32-S3	IoT controller

ESP32-CAM / Camera	Visual monitoring
BME680	Temperature, humidity and air-quality monitoring
MQ-2 or equivalent	Gas/smoke-related monitoring
OLED display	Local status display
Buzzer/Siren	Alarm
SG90 Servo	Prototype smart lock
Air-sampling fan	Biological sample collection
Collection filter	Biological sampling
Molecular biology supplies	DNA extraction and analysis
Power supply	System power
PCB/breadboard	Hardware integration
3D-printed housing	Prototype enclosure

No water-leak sensor will be included.

No PIR sensor will be required for the main security system.

No magnetic door sensor will be required for the main door-detection system.

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18. Team Responsibilities

Computer Engineering — Member 1

- ESP32 programming
- Camera integration
- Sensor integration
- IoT communication
- Hardware architecture

Computer Engineering — Member 2

- AI deployment
- Computer vision
- Door-state detection
- Movement/person detection

- Smart-lock and alarm integration

Software Engineering — Member 1

- Flask backend
- REST API
- Database
- Authentication

Software Engineering — Member 2

- Web dashboard
- User interface
- Camera interface
- Data visualization

Software Engineering — Member 3

- MQTT
- Notification system
- AI/data processing pipeline
- System integration
- Optional machine-learning features

Microbiology/Molecular Biology — Member 1

- Biological target selection
- Air-sampling methodology
- DNA extraction
- Molecular assay development
- Primer selection
- Controls
- Biological result interpretation

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19. Expected Final Demonstration

The final prototype can use a small model house containing:

- Front door
- Room
- Camera
- Environmental sensors

- IoT controller
- Smart lock
- Alarm

During the demonstration:

Scenario 1 — Normal

A known/expected environment is monitored.

No person
Door closed
Normal environment
No biological target

→ SYSTEM SAFE

Scenario 2 — Intrusion

A person approaches the door.

AI detects person
↓
AI detects door opening
↓
AI detects person entering
↓
🚨 SECURITY ALERT

Scenario 3 — Environmental Risk

The system detects prolonged high humidity.

Humidity ↑
↓
Environmental warning
↓
Potential biological-risk condition

Scenario 4 — Biological Result

A collected sample is analyzed using the molecular workflow.

Sample analyzed
↓
Target detected

↓
Result entered/transmitted
↓
🧬 BIOLOGICAL ALERT

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20. Innovation

The project combines four technological domains:

🤖 Artificial Intelligence

For understanding camera data and identifying security events.

🌐 IoT

For connecting sensors, cameras, hardware, server, and user interface.

🧬 Molecular Biology

For investigating biological contamination through molecular analysis.

💻 Software Engineering

For creating the backend, database, dashboard, communication system, and data-processing pipeline.

The major innovation is not simply having these technologies separately, but **connecting them into one system**.

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21. Future Development

Future versions could include:

- More advanced object recognition
- Face recognition
- Multi-camera tracking
- Edge AI processing
- Automatic biological sampling
- Automated molecular detection
- Additional biological targets
- Mobile application

- Cloud deployment
- Machine-learning-based environmental risk prediction
- Automated ventilation
- Smart-home device integration

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22. Conclusion

The **AI + IoT Smart Bio-Security Home System** proposes a new approach to smart-home monitoring by replacing several conventional security sensors with AI-based computer vision while incorporating environmental and molecular biological monitoring.

The camera will provide the primary physical-security information, allowing the system to detect people, movement, and door-state changes without relying on separate motion and door sensors.

IoT technology will connect the hardware to a centralized software platform, while the molecular biology component will investigate potential biological contamination.

The resulting prototype will demonstrate the integration of **Artificial Intelligence, Computer Engineering, Software Engineering, IoT, and Molecular Biology** into a single intelligent home-monitoring system.